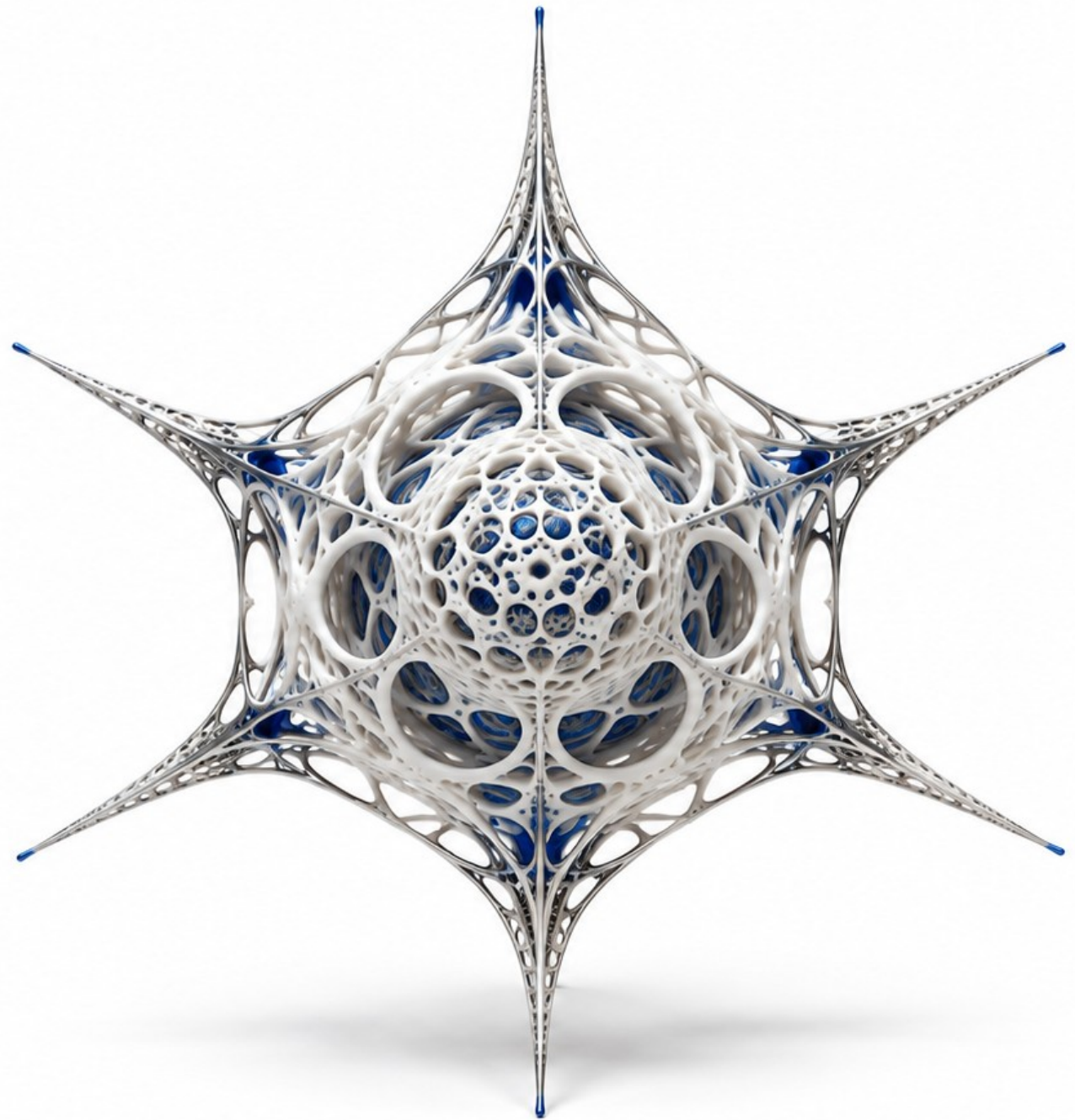


RADIOLARIA OS

Gate 4 Reference

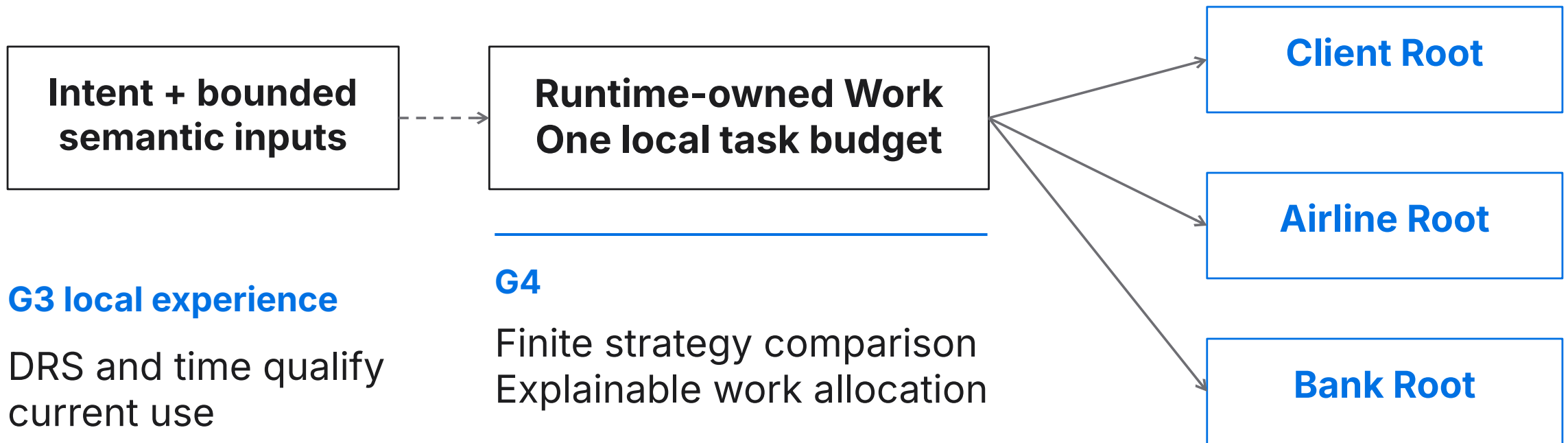
Explicit trade-offs
Bounded work
Independent owners

Owner-approved reduced scope
23 September 2026 / Implementation 5b25944



Mathematics participates in the operating architecture

G4 compares interests and allocates finite local Work. Each Root retains its decision.



Conceptual architecture view. Dashed: advisory input. Solid: evidence for independent review.

Two valid offers express different interests

Controlled Airline records share a EUR 840 ceiling, required baggage and no overnight layover.

offer:0

EUR 790

Standard seat
Window

offer:1

EUR 815

Extra legroom
Aisle

Client: price + seat preference Airline: revenue Bank: budget margin

The declared preference changes the prepared offer

Configured utility units and the finite Nash-product criterion yield different winners.

Profile	Offer	Client U	Airline U	Bank U	Product
PRICE_FIRST	offer:0	0.500	0.790	0.050	0.01975
PRICE_FIRST	offer:1	0.250	0.815	0.025	0.00509375
COMFORT_FIRST	offer:0	0.050	0.790	0.050	0.001975
COMFORT_FIRST	offer:1	0.925	0.815	0.025	0.018846875

PRICE_FIRST selects offer:0

COMFORT_FIRST selects offer:1

Current Root reviews and offer resolution create a prepared hold record
PREPARATION_ONLY_NOT_EXECUTED. No payment or external booking.

Each owner makes a current decision

The recommendation remains offer:0 when Bank consent is missing.

Independent owner	With required consent	Missing Bank consent
Client Root	ACCEPT	ACCEPT
Airline Root	ACCEPT	ACCEPT
Bank Root	ACCEPT	NEEDS_USER

Prepared offer:0

MIXED

No domain continuation

Nine work units remain nine work units

One local task has a fixed cap of nine native Work dispatch units.



1 spent

initial_constraint

7 discretionary checks

Branch limits: 2 minimum, 6 maximum

1 reserved

final_strategy_
consumer

$$9 - 1 - 1 = 7$$

ONE_NATIVE_WORK_DISPATCH_UNIT

An incorrect prediction becomes attributable experience

A controlled prediction is FALSE. The mandatory constraint check returns TRUE.

FALSE

Configured prediction

TRUE

Completed mandatory check

Assessment	Recorded result
Proposal	INCORRECT
Enforcement	ALLOWED_AS_REQUIRED
Execution / task	COMPLETED / COMPLETED

One SPARSE sample through Root-reviewed recording and LocalDRS. No new LLM call.

Experience changes which checks actually run

One source-bound sample changes the seven discretionary checks under PRICE_FIRST.

	Common six checks						Changed check
Cold 4 / 3	0_1 Route	0_2 Baggage	0_3 Seat	1_1 Route	1_2 Baggage	1_3 Seat	0_4 Price
History-informed 3 / 4	0_1 Route	0_2 Baggage	0_3 Seat	1_1 Route	1_2 Baggage	1_3 Seat	1_4 Price

Work IDs use the prefix offer_ (for example offer_0_4)

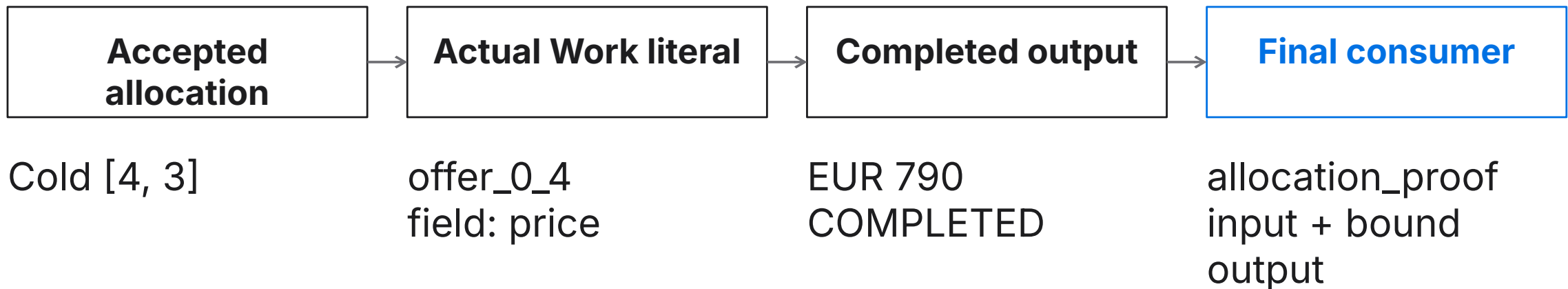
z offer0: 0.425 to 0.4171875

z offer1: 0.42125

Raw prior applies once. Both lanes prepare offer:0.

Allocated work reaches the final consumer

The native Work input carries the allocation proof. Its output binds the same allocation.



Coherent 6 / 1 substitution

REFUSED before changed execution

Original source basis disagrees. The intact allocation is the passing neighbour.

Current context remains binding

Valid old records remain subject to the original budget, expiry and current history checks.

Boundary	Genuine positive	Refused continuation
Original budget	Seven optional checks plus mandatory endpoints	Fresh report cannot erase spent work
Original expiry	Just before expiry: prepared result	At boundary or after: REFUSED
Retained history	Current discovery and Root-approved descent	Expired retained input: REFUSED

The saved result can be traced and checked

Numerical references, recorded native work and supplied verification have distinct scopes.

Evidence layer	Recorded scope	Result
G41 numerical	110 pure-math tests	Recorded reference
G43 native	3 tasks × 9 units = 27	Actual Work + consumption
G44 focused	33 PASS / 1 FAIL	Accepted mode-test limitation
G44 / G45 supplied	3 identical complete outputs	Saved relationships checked

Supplied checks require the trusted installed code and matching recorded environment.

Full native replay is unsupported. No new LLM or Work calls occur in the named replay scope.

Reference accepted, deeper mathematics deferred

Preferences change the prepared offer. Experience changes work allocation.

ACCEPTED REFERENCE

Finite multi-Root comparison
Bounded allocation of actual Work
Independent owner decisions

DEFERRED PROGRAMME

Full H-I-J after Gate 6 publication
Separate future scope decision
LGT excluded

Reference engineering accepted / presentation prepared

[Technical appendix](#)

[XML Reader](#)

[Reader index](#)

[Pinned implementation](#)

[Verification guide](#)

Repository access required. Document links are staged publication targets on main.